Where, LHS is $(3+4)(3+2)=7\times5=35$

RHS is $(3\times3)+3(4+2)+(4\times2)=9+(3\times6)+8$

$=9+18+8=35$

Therefore, LHS = RHS.

In the similar way as explained above, let us check for another set of four regions as shown in the *Fig. 3.9*.

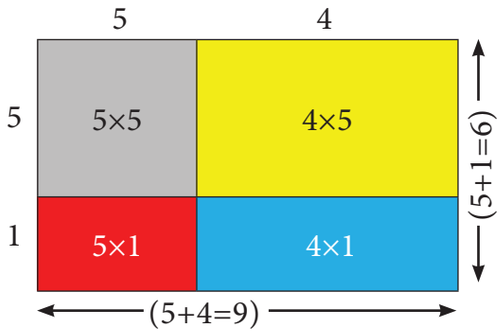


Fig. 3.9

By observing *Fig. 3.9*, we can note that,

Area of the bigger rectangle = Area of a square (Grey) + Area of Three rectangles

$$(5+4)\,(5+1) = (5\times5) + (5\times1) + (5\times4) + (1\times4)$$

$$(5+4)\,(5+1) = (5\times5) + 5(1+4) + (1\times4) \qquad ...(2)$$

Where, LHS is $(5+1)(5+4)=6\times9=54$

RHS is $5^2+5(1+4)+(1\times4)=25+(5\times5)+4$

$=25+25+4=54$

Therefore, LHS = RHS.

Thus, equation (1) and (2) is true for given set of any three values.

By generalising those three values as 'x', 'a' and 'b'

we get, $(x+a)(x+b)=(x\times x)+x(a+b)+(a\times b)$

That is, $(x+a)(x+b)=x^2+x(a+b)+ab$

Hence, $(x+a)(x+b)=x^2+x(a+b)+ab$ is an identity.

Now, let us prove this identity, geometrically.

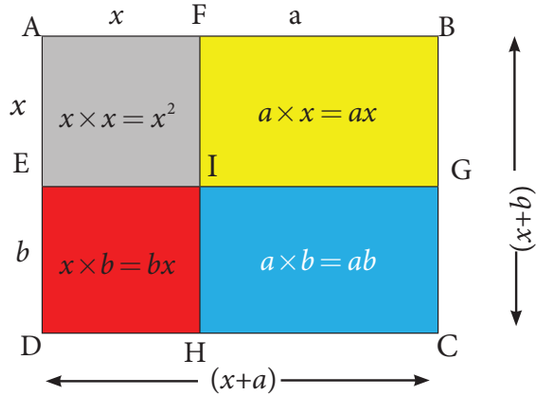


Fig. 3.10

Note

We know that,

Area of rectangle
$=length\times breadth$
$=l\times b$

Also, area of square
$=side\times side$
$=a\times a=a^2$

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